Zhanpeng Zhou Ph.D. Candidate

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Education

School of Computer Science, Shanghai Jiao Tong University — Shanghai, China
Ph.D. in Computer Science and Engineering — 2022 - 2027 *(expected)*

- Advisor: Prof. Junchi Yan
- Research area: **Science of Large Language Models**, Theory of Deep Learning

UM-SJTU Joint Institute, Shanghai Jiao Tong University — Shanghai, China
B.E. in Electrical and Computer Engineering — 2018 - 2022

- Minor in Data Science.

Experiences

Tencent Hunyuan Pretraining Group. | Beijing, China — 2026/05 - present

- Research Intern, with **Qingyun** Talent Program.

ByteDance Seed Edge. | Beijing, China — 2025/12 - 2026/05

- Research Intern, focusing on **optimizer and hyperparameter transfer** in pretraining.

MiniMax Inc. | Shanghai, China — 2025/04 - 2025/11

- Research Intern, contributing to **MiniMax-M2 model.**

National Institute of Informatics. | Tokyo, Japan — 2023/09 - 2024/03

- Research Intern, advised by Prof. Mahito Sugiyama.

Publications (Selected)

(* indicates equal contributions; †indicates correspondence.)

1. Jianhao Huang*, **Zhanpeng Zhou***, Renqiu Xia, Baharan Mirzasoleiman, Weijie Su, Wei Huang. How Transformers Learn to Plan via Multi-Token Prediction. **In Submission.**
2. **Zhanpeng Zhou***†, Yuhan Sun*, Bingrui Li, Jinbo Wang, Huaijin Wu, Lei Wu, Junchi Yan. Sharp-to-Flat Early, Noise-Governed Late: Local Geometry in Language Model Pre-Training. **In Submission.**
3. Tongtian Zhu, Tianyu Zhang, Mingze Wang, **Zhanpeng Zhou**†, Can Wang. On the Surprising Effectiveness of a Single Global Merging in Decentralized Learning. **ICLR 2026 (Oral).**
4. Tongcheng Zhang*, **Zhanpeng Zhou***, Mingze Wang, Andi Han, Wei Huang, Taiji Suzuki, Junchi Yan. On the Learning Dynamics of Two-Layer Networks with Label Noise SGD. **AAAI 2026 (Oral).**
5. Jinbo Wang*, Mingze Wang*, **Zhanpeng Zhou***, Junchi Yan, Weinan E, Lei Wu. The Sharpness Disparity Principle in Transformers for Accelerating Language Model Pre-Training. **ICML 2025.**
6. Andi Han*†, Wei Huang*†, **Zhanpeng Zhou***†, Gang Niu, Wuyang Chen, Junchi Yan, Akiko Takeda, Taiji Suzuki. On the Role of Label Noise in the Feature Learning Process. **ICML 2025.**
7. **Zhanpeng Zhou***†, Mingze Wang*, Yuchen Mao, Bingrui Li, Junchi Yan†. Sharpness-Aware Minimization Efficiently Selects Flatter Minima Late in Training. **ICLR 2025 (Spotlight).**
8. **Zhanpeng Zhou***, Zijun Chen*, Yilan Chen, Bo Zhang, Junchi Yan. On the Emergence of Cross-Task Linearity in the Pretraining-Finetuning Paradigm. **ICML 2024.**
9. **Zhanpeng Zhou***, Wen Shen*, Huixin Chen*, Ling Tang, Quanshi Zhang. Batch Normalization Is Blind to the First and Second Derivatives of the Loss. **AAAI 2024 (Oral).**
10. **Zhanpeng Zhou**, Yongyi Yang, Xiaojiang Yang, Junchi Yan, Wei Hu. Going Beyond Linear Mode Connectivity: The Layerwise Linear Feature Connectivity. **NeurIPS 2023.**

Publications (Others)

(* indicates equal contributions; †indicates correspondence.)

1. Junchi Yan, Yiting Chen, Zongwei Huo, **Zhanpeng Zhou**, Theodore Papamarkou, Cesare Alippi. Iterative Feature Merging for Neural Network Compression and Parameter-Efficient Fine-Tuning. **In Submission.**
2. Bingrui Li, Jiaxin Wen, **Zhanpeng Zhou**, Jun Zhu, Jianfei Chen. Efficient Hyperparameter Tuning via Trajectory Invariance Principle. **In Submission.**
3. Qibing Ren, Xinhao Song, Ke Fan, Lijun Li, **Zhanpeng Zhou**, Gongshen Liu, Junchi Yan, Lizhuang Ma, Jing Shao. More Sail than Ballast: Addressing Harmful Knowledge Leakage in the Expansive Reasoning Space of LRMs. **ICML 2026.**
4. Jinbo Wang, Binghui Li, **Zhanpeng Zhou**, Mingze Wang, Yuxuan Sun, Jiaqi Zhang, Xunliang Cai, Lei Wu. Fast Catch-Up, Late Switching: Optimal Batch Size Scheduling via Functional Scaling Laws. **ICLR 2026.**
5. Xuan Cui, Yunfei Zhao, Huiyue Li, Run Zeng, Jinrui Qian, Wei Duan†, Bo Liu†, **Zhanpeng Zhou†**. IGU-LoRA: Adaptive Rank Allocation via Integrated Gradients and Uncertainty-Aware Scoring. **ICLR 2026.**
6. Huaijin Wu, Bingrui Li, Yebin Yang, Yi Tu, **Zhanpeng Zhou**, Jianfei Chen, Junchi Yan. Achieving Low-Bit Muon Through Subspace Preservation and Grid Quantization. **ICLR 2026.**
7. Zijun Chen*, **Zhanpeng Zhou***, Bo Zhang, Weinan Zhang, Xi Sun, Junchi Yan. SE-Merging: A Self-Enhanced Approach for Dynamic Model Merging. **IJCNN 2025.**
8. **Zhanpeng Zhou**, Yongyi Yang, Mahito Sugiyama, Junchi Yan. New Evidence of the Two-Phase Learning Dynamics of Neural Networks. **ICML 2025 Workshop.**
9. **Zhanpeng Zhou**, Yongyi Yang, Jie Ren, Mahito Sugiyama, Junchi Yan. On the Cone Effect in the Learning Dynamics. **ICLR 2025 Workshop.**
10. Bingrui Li, Wei Huang, Andi Han, **Zhanpeng Zhou**, Taiji Suzuki, Jun Zhu, Jianfei Chen. On the Optimization and Generalization of Two-layer Transformers with Sign Gradient Descent. **ICLR 2025 (Spotlight).**
11. Yiting Chen, **Zhanpeng Zhou**, Junchi Yan. Going Beyond Neural Network Feature Similarity: The Network Feature Complexity and Its Interpretation Using Category Theory. **ICLR 2024.**
12. Jie Ren, **Zhanpeng Zhou**, Qirui Chen, Quanshi Zhang. Can We Faithfully Represent Absence States to Compute Shapley Values on a DNN? **ICLR 2023.**

Projects

PMuon | Preconditioning Muon with Gradient Covariance Before the Polar Step
- SOTA on Track 3 Optimization of modded-nanogpt, tying NorMuon.

Awards

- **Top Reviewer Award**, NeurIPS 2025 — 2025/10
- **National Scholarship (top 0.2%)**, Ministry of Education — 2024/11
- **Top Internship Evaluation**, National Institute of Informatics — 2024/03
- **Outstanding Graduate Student**, Shanghai Jiao Tong University, — 2022/05
- **Yu Liming Scholarship**, Shanghai Jiao Tong University — 2021/11
- **John Wu & Jane Sun Scholarship**, Shanghai Jiao Tong University — 2020/11
- **Best Technology Award in Summer Expo**, Shanghai Jiao Tong University — 2019/08

Services

Area Chair: *CPAL '26, ICML '26 FoGen, ICLR '26 DeLTa*
Conference Reviewer: *ICML ('22-25), NeurIPS ('22-25), ICLR ('24-26)*
Journal Reviewer: *T-PAMI, Intelligent Computing (Science Partner)*